

ASSIGNMENT NO.4: Working with Multiple Tables and Joins

Questions: Q1) Create a MySQL database named SeniorCollegeDB and create the following tables: Student Table Column Name Data Type StudentID INT Constraint StudentName VARCHAR(50) PRIMARY KEY Column Name Data Type Constraint City VARCHAR(30) Course Table Column Name Data Type Constraint CourseID INT PRIMARY KEY CourseName VARCHAR(50) Fees DECIMAL(10,2) Enrollment Table Column Name Data Type Constraint EnrollmentID INT PRIMARY KEY StudentID INT FOREIGN KEY CourseID INT FOREIGN KEY EnrollmentDate DATE

ANS:

-- Create Database

```
CREATE DATABASE SeniorCollegeDB;
```

```
USE SeniorCollegeDB;
```

-- Create Student Table

```
CREATE TABLE Student (  
    StudentID INT PRIMARY KEY,  
    StudentName VARCHAR(50),  
    City VARCHAR(30)  
);
```

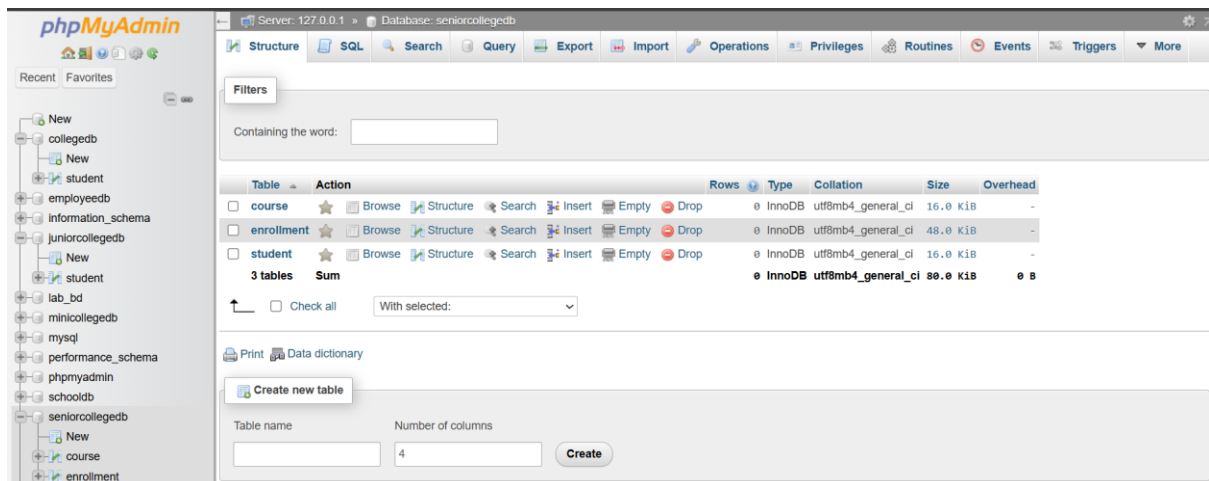
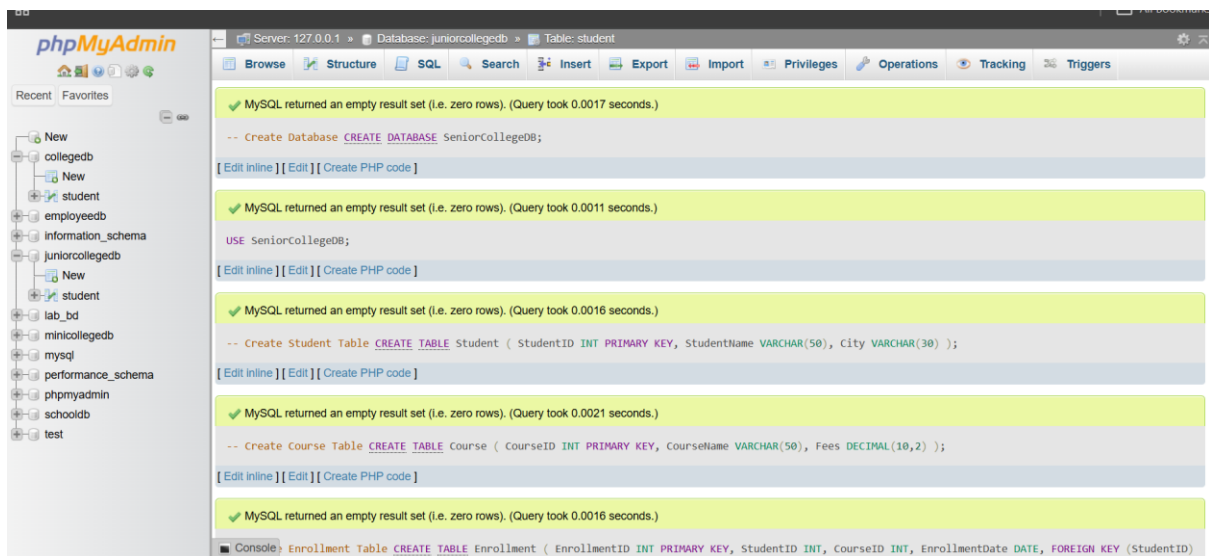
-- Create Course Table

```
CREATE TABLE Course (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(50),  
    Fees DECIMAL(10,2)  
);
```

-- Create Enrollment Table

```
CREATE TABLE Enrollment (  
    EnrollmentID INT PRIMARY KEY,  
    StudentID INT FOREIGN KEY REFERENCES Student(StudentID),  
    CourseID INT FOREIGN KEY REFERENCES Course(CourseID),  
    EnrollmentDate DATE
```

EnrollmentID INT PRIMARY KEY,
 StudentID INT,
 CourseID INT,
 EnrollmentDate DATE,
 FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
 FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
);



Q2) Insert at least 6 records into the Student table, 4 records into the Course table, and 5 records into the Enrollment table. Make sure:

Requirement
At least one student should not have enrollment
At least one course should not have any enrolled student

ANS:

-- Insert into Student table

INSERT INTO Student VALUES

(1, 'Alice', 'NY'),

(2, 'Bob', 'LA'),

(3, 'Charlie', 'SF'),

(4, 'Dave', 'CHI'),

(5, 'Eve', 'MIA'),

(6, 'Frank', 'SEA');

-- Insert into Course table

INSERT INTO Course VALUES

(101, 'Math', 500.00),

(102, 'Science', 600.00),

(103, 'Art', 300.00),

(104, 'History', 400.00);

-- Insert into Enrollment table

INSERT INTO Enrollment VALUES

(1001, 1, 101, '2023-01-01'),

(1002, 2, 101, '2023-01-02'),

(1003, 3, 102, '2023-01-03'),

(1004, 4, 103, '2023-01-04'),

(1005, 5, 103, '2023-01-05');

phpMyAdmin

Server: 127.0.0.1 Database: seniorcollegedb

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers More

Show query box

6 rows inserted. (Query took 0.0039 seconds.)

```
-- Insert into Student table INSERT INTO Student VALUES (1, 'Alice', 'NY'), (2, 'Bob', 'LA'), (3, 'Charlie', 'SF'), (4, 'Dave', 'CHI'), (5, 'Eve', 'MIA'), (6, 'Frank', 'SEA');
```

[Edit inline] [Edit] [Create PHP code]

4 rows inserted. (Query took 0.0061 seconds.)

```
-- Insert into Course table INSERT INTO Course VALUES (101, 'Math', 500.00), (102, 'Science', 600.00), (103, 'Art', 300.00), (104, 'History', 400.00);
```

[Edit inline] [Edit] [Create PHP code]

5 rows inserted. (Query took 0.0016 seconds.)

```
-- Insert into Enrollment table INSERT INTO Enrollment VALUES (1001, 1, 101, '2023-01-01'), (1002, 2, 101, '2023-01-02'), (1003, 3, 102, '2023-01-03'), (1004, 4, 103, '2023-01-04'), (1005, 5, 103, '2023-01-05');
```

[Edit inline] [Edit] [Create PHP code]

Showing rows 0 - 3 (4 total, Query took 0.0016 seconds.)

SELECT * FROM `course`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		CourseID	CourseName	Fees				
☐	Edit	☐	Copy	☐	Delete	101	Math	500.00
☐	Edit	☐	Copy	☐	Delete	102	Science	600.00
☐	Edit	☐	Copy	☐	Delete	103	Art	300.00
☐	Edit	☐	Copy	☐	Delete	104	History	400.00

☐ Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

SELECT * FROM `enrollment`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		EnrollmentID	StudentID	CourseID	EnrollmentDate				
☐	Edit	☐	Copy	☐	Delete	1001	1	101	2023-01-01
☐	Edit	☐	Copy	☐	Delete	1002	2	101	2023-01-02
☐	Edit	☐	Copy	☐	Delete	1003	3	102	2023-01-03
☐	Edit	☐	Copy	☐	Delete	1004	4	103	2023-01-04
☐	Edit	☐	Copy	☐	Delete	1005	5	103	2023-01-05

☐ Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

SELECT * FROM `student`

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		StudentID	StudentName	City				
☐	Edit	☐	Copy	☐	Delete	1	Alice	NY
☐	Edit	☐	Copy	☐	Delete	2	Bob	LA
☐	Edit	☐	Copy	☐	Delete	3	Charlie	SF
☐	Edit	☐	Copy	☐	Delete	4	Dave	CHI
☐	Edit	☐	Copy	☐	Delete	5	Eve	MIA
☐	Edit	☐	Copy	☐	Delete	6	Frank	SEA

☐ Check all | With selected: ☐ Edit ☐ Copy ☐ Delete ☐ Export

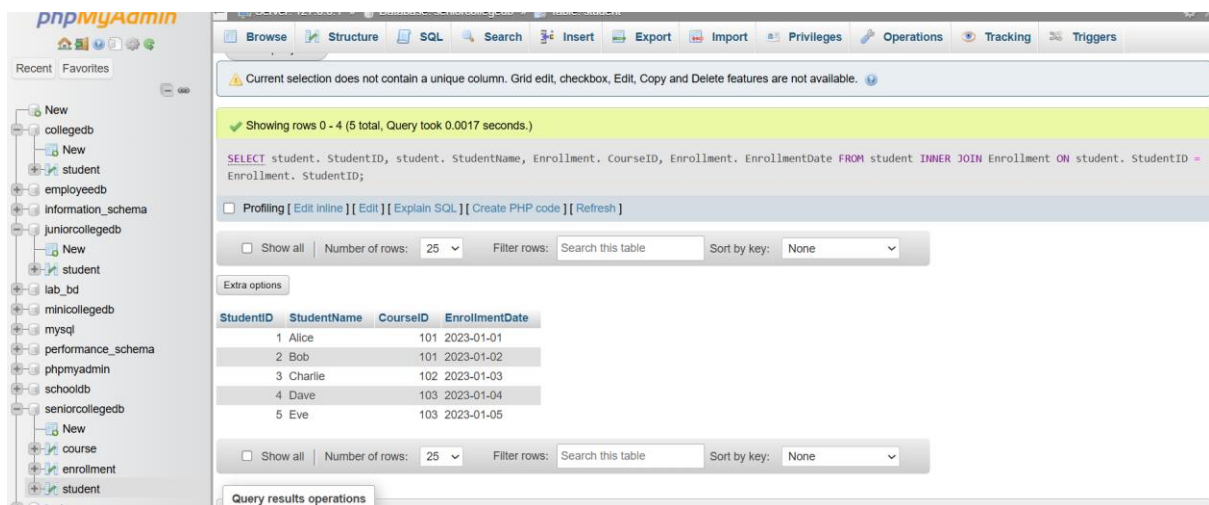
☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Q3) Write a MySQL query using INNER JOIN to display students who have enrolled in courses. The output should include:

Column
StudentID
StudentName
CourseID
EnrollmentDate

ANS:

```
SELECT student. StudentID,
student. StudentName, Enrollment. CourseID,
Enrollment. EnrollmentDate
FROM student
INNER JOIN Enrollment ON
student. StudentID =
Enrollment. StudentID;
```



Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 4 (5 total, Query took 0.0017 seconds.)

```
SELECT student. StudentID, student. StudentName, Enrollment. CourseID, Enrollment. EnrollmentDate FROM student INNER JOIN Enrollment ON student. StudentID = Enrollment. StudentID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

StudentID	StudentName	CourseID	EnrollmentDate
1	Alice	101	2023-01-01
2	Bob	101	2023-01-02
3	Charlie	102	2023-01-03
4	Dave	103	2023-01-04
5	Eve	103	2023-01-05

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Q4) Write a MySQL query to display student name, city, course name, fees, and enrollment date using multiple tables. Use:

Tables Required
Student
Course
Enrollment

ANS:

```
SELECT student. StudentName,  
student. city,  
Course. CourseName,  
Course. Fees,  
Enrollment. EnrollmentDate  
FROM student  
JOIN Enrollment ON  
student. StudentID =  
Enrollment. StudentID  
JOIN Course ON Enrollment. CourseID  
= Course. CourseID;
```

The screenshot shows a database management interface. On the left is a sidebar with a tree view of databases and tables. The main area displays a SQL query and its results. The query is:

```
SELECT student. StudentName, student. city, Course. CourseName, Course. Fees, Enrollment. EnrollmentDate FROM student JOIN Enrollment ON student. StudentID = Enrollment. StudentID JOIN Course ON Enrollment. CourseID = Course. CourseID;
```

Below the query, there are controls for showing all rows, number of rows (25), filter rows, and sort by key (None). The results table shows 5 rows of data:

StudentName	city	CourseName	Fees	EnrollmentDate
Alice	NY	Math	500.00	2023-01-01
Bob	LA	Math	500.00	2023-01-02
Charlie	SF	Science	600.00	2023-01-03
Dave	CHI	Art	300.00	2023-01-04
Eve	MIA	Art	300.00	2023-01-05

Q5) Write a MySQL query using LEFT JOIN to display all students, including those who have not enrolled in any course. The output should include:

Column
StudentID
StudentName
City
CourseID
EnrollmentDate

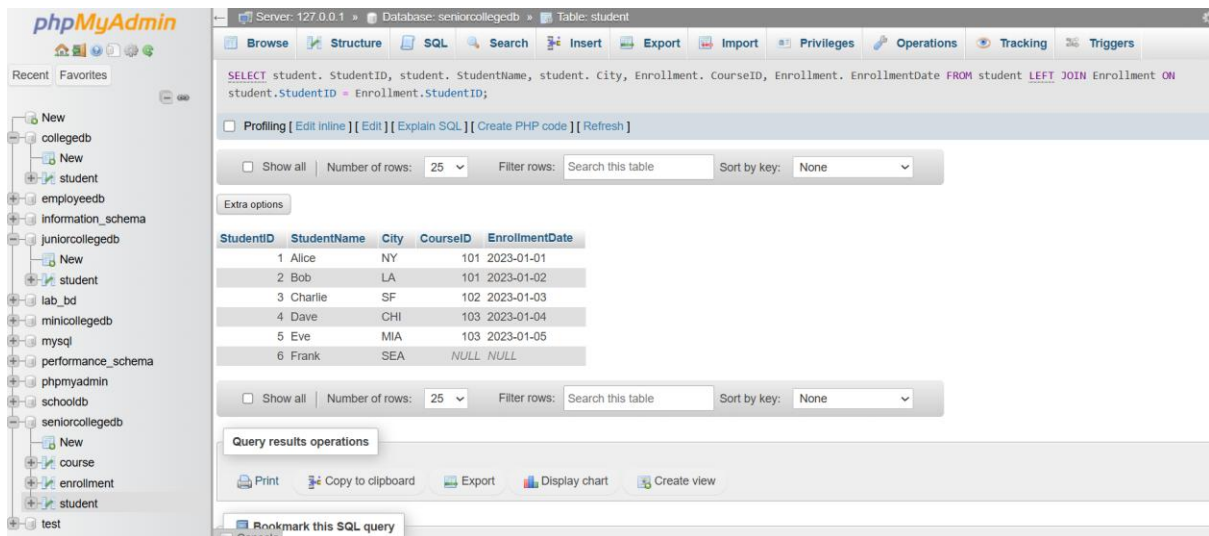
ANS:

```
SELECT student. StudentID, student. StudentName,
```

```

student. City,
Enrollment. CourseID,
Enrollment. EnrollmentDate
FROM
student
LEFT JOIN Enrollment
ON student.StudentID = Enrollment.StudentID;

```



Q6) Write a MySQL query using RIGHT JOIN to display all courses, including courses that have no enrolled students. The output should include:

Column
StudentName
CourseID
CourseName
Fees

ANS:

```

SELECT student.StudentName,
Course. CourseID, Course. CourseName,
Course. Fees
FROM student
JOIN Enrollment ON

```

student.StudentID = Enrollment.StudentID

RIGHT JOIN Course ON

Enrollment.CourseID = Course.CourseID;

phpMyAdmin

Server: 127.0.0.1 » Database: seniorcollegedb » Table: student

Current selection does not contain a unique column. Grid edit, checkbox, Edit, Copy and Delete features are not available.

Showing rows 0 - 5 (6 total, Query took 0.0017 seconds.)

```
SELECT student.StudentName, Course.CourseID, Course.CourseName, Course.Fees FROM student JOIN Enrollment RIGHT JOIN Course ON Enrollment.CourseID = Course.CourseID;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

StudentName	CourseID	CourseName	Fees
Alice	101	Math	500.00
Bob	101	Math	500.00
Charlie	102	Science	600.00
Dave	103	Art	300.00
Eve	103	Art	300.00
NULL	104	History	400.00

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations